

COMS5023A

Lecturer: Dr Alton Dewa

E-mail: alton.dewa@wits.ac.za

SQL Data Manipulation Assessment Questions

Total Marks: 50

Question 1 [6 marks]

Write the SQL command to create the students table with the specified attributes and data types (student_num, student_lname, student_fname, enrollment_date, age).

ATTRIBUTE (FIELD NAME)	DATA TYPE
student_num	CHAR(3)
student_lname	VARCHAR(25)
student_fname	VARCHAR(25)
enrollment_date	DATE
age	INTEGER

Describe the table by executing the command

DESC students;

Your output should be as shown in the screenshot:

```
mysql> DESC students;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| student_num | char(3) | YES | | NULL | |
| student_lname | varchar(25) | YES | | NULL | |
| student_fname | varchar(25) | YES | | NULL | |
| enrollment_date | date | YES | | NULL | |
| age | int | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

Question 2 [8 marks]

Write SQL statements to insert all six student records provided in the exercise into the students table. Your output should be as shown in the screenshot

student_num	student_lname	student_fname	enrollment_date	age
101	Mokoena	Thabo	2019-05-23	18
102	Shilubana	Ndivhuwo	2018-08-13	20
103	Baloyi	Tsakani	2017-11-09	19
104	Mgangatho	Lukhanyo	2020-03-17	17
105	Mofokeng	Lerato	2018-12-05	21
106	Mnguni	Nomsa	2018-08-13	18

Question 3 [4 marks]

Write the SQL command to insert a new student with student_num 054 and age 7 only.

Your output should be as shown in the screenshot

student_num	student_lname	student_fname	enrollment_date	age
101	Mokoena	Thabo	2019-05-23	18
102	Shilubana	Ndivhuwo	2018-08-13	20
103	Baloyi	Tsakani	2017-11-09	19
104	Mgangatho	Lukhanyo	2020-03-17	17
105	Mofokeng	Lerato	2018-12-05	21
106	Mnguni	Nomsa	2018-08-13	18
054	NULL	NULL	NULL	7

Question 4 [4 marks]

Write the SQL command to insert a new student with student_num 076, student_lname Vukosi, and enrollment_date 2021-09-09. Your output should be as shown in the screenshot

	student_num	student_lname	student_fname	enrollment_date	age
1	101	Mokoena	Thabo	2019-05-23	18
2	102	Shilubana	Ndivhuwo	2018-08-13	20
3	103	Baloyi	Tsakani	2017-11-09	19
4	104	Mgangatho	Lukhanyo	2020-03-17	17
5	105	Mofokeng	Lerato	2018-12-05	21
6	106	Mnguni	Nomsa	2018-08-13	18
7	054	NULL	NULL	NULL	7
8	076	Vukosi	NULL	2021-09-09	NULL

Question 5 [2 marks]

Write the SQL command to display all records from the students table.

Question 6 [3 marks]

Write the SQL command to display only the student_num and age columns for all students. .

Your output should be as shown in the screenshot

	student_num	age
1	101	18
2	102	20
3	103	19
4	104	17
5	105	21
6	106	18
7	054	7
8	076	NULL

Question 7 [5 marks]

Write the SQL command to update the first name of the student with student_num 104 to Mark.

Question 8 [6 marks]

Write the SQL command to update the age of all students enrolled before 2019 to 22. Your output should be as shown in the screenshot

	student_num	student_lname	student_fname	enrollment_date	age
1	101	Mokoena	Thabo	2019-05-23	18
2	102	Shilubana	Ndivhuwo	2018-08-13	22
3	103	Baloyi	Tsakani	2017-11-09	22
4	104	Mgangatho	Lukhanyo	2020-03-17	17
5	105	Mofokeng	Lerato	2018-12-05	22
6	106	Mnguni	Nomsa	2018-08-13	22
7	054	NULL	NULL	NULL	7
8	076	Vukosi	NULL	2021-09-09	NULL

Question 9 [5 marks]

Write the SQL command to delete all records where age equals 18. Your output should be as shown in the screenshot

	student_num	student_lname	student_fname	enrollment_date	age
1	102	Shilubana	Ndivhuwo	2018-08-13	22
2	103	Baloyi	Tsakani	2017-11-09	22
3	104	Mgangatho	Lukhanyo	2020-03-17	17
4	105	Mofokeng	Lerato	2018-12-05	22
5	106	Mnguni	Nomsa	2018-08-13	22
6	054	NULL	NULL	NULL	7
7	076	Vukosi	NULL	2021-09-09	NULL

Question 10 [7 marks]

Write the SQL command to delete all rows from the students table without deleting the table structure. . Your output should be as shown in the screenshot

student_num	student_lname	student_fname	enrollment_date	age
-------------	---------------	---------------	-----------------	-----